

Conventional NIRS calibration

external pipeline search

many preprocessing
pipelines

CV selects a
complete pipeline

final model depends
on pipeline choice

external choices are
hard to audit

refitting repeats
the search

Analytical bottleneck

cost, variance, opacity

high calibration
cost

winner's curse
& instability

deployment
complexity

poor traceability

limited routine
usability

Operator-adaptive calibration

model-internal & auditable

compact linear
operator bank

selection inside the
calibration model

original-wavelength
coefficients

selected operators
logged

fast refit of one
linear model

AOM reframes preprocessing selection as a model-internal, auditable part of analytical calibration.